An EMG-to-Force Processing Approach for Estimating in Vivo Hip Muscle Forces in Normal Human Walking

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***Abstract*— The force of a single muscle is not directly measurable without invasive methods. Yet invasive techniques are not appropriate for clinical use, thus a non-invasive technique that combined the electromyographic (EMG) signal and a neuromuscular model was developed to determine in vivo active muscle forces at the hip. The EMG-to-force processing (EFP) model included active and passive moment components, and the net EFP moment was compared with the hip moment obtained with standard inverse dynamics techniques ("gold standard"). The two methods were compared at percent gait cycle intervals, and the correlation coefficient between methods was excellent (r^2 = 91). The closeness of fit confirms that the approach is a reasonable approximation of net moment and, possibly, individual muscle forces. The greatest estimated hip force was produced by a hip abductor. A novel finding was that the hip adductors did not behave a single synergistic group. The Adductor Magnus synergistically assisted other hip extensors, and produced forces that were out-of-phase with the other hip adductor forces. Rectus Femoris was only active during hip flexion (not knee extension).**

***Index Terms*— Hip, muscle force, neuromuscular model, normal gait.**

I. Introduction

THE determination of muscle forces during movement is essential for a complete understanding of the underlying neural control of movement. Yet direct measurement of muscle force is at best impractical and in most cases not possible. Thus, biomechanical research has been directed towards developing accurate indirect muscle force estimation techniques.

Most skeletal joints are actuated by more muscles than are needed to perform basic movements. Yet the reason for the overabundance of available muscles at each joint is not obvious. From a structural perspective we are left with statistical indeterminacy (more unknown forces than solution set of equations), where assessment of muscle role is sometimes resolved by resorting to some form of optimization criteria. A single cost function is typically minimized or maximized (e.g., muscle endurance, muscle stress, metabolic cost) as kinematic data is combined with certain objective function assumptions [1]–[4]. Our insights into gait mechanics and efficiency have been advanced as a result of these studies. However two aspects related to optimization research remain limitations. First, obtaining a solution can be very time-intensive [5]. A more important issue that cannot be circumvented is that the determination of objective functions remains subjective, as several solution sets may satisfy the stated criteria [1]. Another approach that indirectly estimates muscle forces is inverse dynamics. Here the joint moments are calculated from kinetic and kinematic data by means of a quasi-rigid-body- segment model. Again, the redundancy of the musculoskeletal system limits this approach, notably by the inability to correctly distribute predicted forces to the individual tissues in and around the joints. Thus synergistic muscles that act at a joint are grouped, reducing the number of unknowns. The residual (net) moments associated with movement at a joint can be established, and muscle activity inferred from the magnitude and sign of the joint moment. If musculoskeletal geometry is included, muscle forces could (in theory) be estimated from the joint moments. Presumably a significant portion of the residual moment is due to muscle activity, with passive elements such as ligaments and other connective tissues also contributing. However, single muscle forces cannot generally be determined and the interpretation of results is confounded in the presence of co-contraction of antagonistic muscle(s) [6], [25]. Electromyography (EMG) is a standard tool in movement analysis. EMG magnitude is not directly correlated with muscle force; however, a technique relating EMG relative intensity to muscle force production has emerged [EMG-to-force processing (EFP)]. Reasonable estimates of muscle forces predicted by EFP have been shown in animal [7], [8] and human trials [9]–[14]. Power produced by individual muscles has been reported at the ankle, only [15]. It is the purpose of this study to use an enhanced EFP technique to estimate the forces generated by individual hip muscles during comfortable speed walking in neurologically-intact adults.

METHODS

SUBJECTS

A convenience sample of twenty neurologically- intact adult males was recruited to participate in the study. Two subjects did not participate in repeat gait trials, and their data were not further examined. The 18 remaining study participants had a mean age of 29 $\pm$ 3.1 years (range 24–34 years), and a mean mass of 76.0 kg ($\pm$ 6.75 kg). The mean walking velocity for these subjects was 81.9 ± 3.7 meters per minute. Subjects consented to participate following explanation of the procedure and review of the informed consent, as approved by

Manuscript received December 4, 2014; revised July 1, 2015; accepted July 25, 2015. Date of publication October 12, 2016; date of current version August 7, 2017.

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Digital Object Identifier 10.1109/TNSRE.2016.2613021

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the Institute Review Board, and signed the Rights of Human Subjects form.

METHODS

Kinematic, kinetic, footswitch and EMG data were simultaneous acquired while study participants performed self-selected speed walking. Each trial consisted of traversing the middle 6 m of a level, 12-m walkway. Due to the comprehensive number of muscles examined each study participant performed two walking trials, with data selected from each representative trial. The examined gait parameters were highly consistent across trials for neurologically-intact adults and the use of multiple sessions did not confound the results.

Kinetics plus kinematics

Subjects wore appropriate clothing, consisting of running tights and a comfortable shirt. Footswitches, similar to insoles, were placed into the study participant's shoes. Each footswitch had compression-closing contacts located at the heel, first and fifth metatarsal heads, and great toe. Round reflective markers were attached to the skin with double-sided tape using anatomic landmarks [16] in order to determine joint centers of rotation. Motion data was sampled at 100 Hz with a twelve-camera digital motion capture system (Model T20, Vicon Motion Systems, Oxford, U.K.). Calibration was performed prior to gait trials. Ground reaction force data was collected at 1000 Hz with paired, in-series force platforms (Type OR6–5, Advanced Medical Technology Incorporated, Newton, MA, USA). Forces and moments in three directions were acquired, and motion and ground reaction force data were synchronized. Subject body weights were obtained. Individual segment lengths were determined from marker locations. Mass and center of mass location for each segment were established [17]. After study participants were weighed (using the force platform) they performed level, self-selected speed walking. Sufficient trials were performed prior to the beginning of data acquisition to allow the study participants to become acclimated to the test environment. While data was acquired from both legs, only data (motion, force, EMG) from the right leg was used for statistical analysis. Normal gait is essentially symmetric and thus no loss of generalizability of findings should result. Walking speed was determined by the time required to traverse the known distance. Walking velocities were plotted for each trial, and the five trials most closely representing typical walking speed for each study participant were averaged. Marker coordinates were bi-directionally smoothed with a fourth-order Butterworth filter with an effective cutoff frequency of 6 Hz [18]. Linear velocities and accelerations, angular position, velocity, and acceleration of segments were determined using software developed in-house. Joint position data were input to the Force Processor as well (*below*). Ground reaction force data was smoothed with a five point moving average. Location of the center of pressure was determined [18]. Motion and ground reaction force data were temporally matched. Finally, moments in three dimensions were obtained using Newtonian equations.

EMG-To-Force Processing—Force Processor

Using the software OpenSim [19] a generic model of the human musculoskeletal geometry was matched to each study participant's anthropometry using anatomical landmarks and functional joint centers as a reference. The model consisted of three-dimensional representations of the bones and muscle-tendon paths, kinematic models of the ankle, knee and hip, and a nominal biomechanical model of each musculotendinous unit (MTU) [20]. The isometric force generating properties of a specific MTU were derived by scaling a generic, Hill-based model [21]. Four parameters scaled the model to represent a specific muscle-tendon complex, and were the input for estimating each study participant's unique MTU forces during self- selected speed level walking. The four parameters—peak isometric force, optimal muscle fiber length, pennation angle, and tendon slack length—were derived from data collected in anatomical studies. Peak isometric force was based on the physiological cross-sectional area of the muscle [22]. Optimal muscle-fiber length was the fiber length at which active muscle force peaks [23], [24]. Pennation angle represents the angle between the tendon long axis and the muscle fibers. Values for optimal fiber length and pennation angle were adapted from Wickiewicz [22]. For muscles not otherwise described the values of fiber length and pennation angle reported by Friedrich [25] were used. Tendon slack length [26] was the MTU length beyond which force develops on elongation. The active moment contribution of each muscle was the product of its force generation and moment arm magnitude. Moment arms were calculated using the partial velocity method described by Yamaguchi [27]. The generic model [28] accounted for the static properties of both muscle [23] and tendon [29]. The force processor was scaled in a non-linear fashion to reflect the force-velocity characteristics of muscle contraction. Further scaling of the model was made to reflect muscle contraction histories that indicate storage and release of energy in tendons and muscle cross-bridges. Muscle force was augmented when concentric contractions immediately followed eccentric muscle contraction [26], [30], [31].

EMG-To-Force Processing—Dynamic Electromyography (EMG)

As part of multiple-muscle studies the electromyographic (EMG) activity of 14 muscles crossing the hip was examined. Muscles tested included the Gluteus Maximus, Adductor Magnus, Semitendinosis, Semimembranosis, long head of the Biceps Femoris, Adductor Longus, Adductor Brevis, Gracilis, Gluteus Medius, Gluteus Minimus, Tensor Fascia Lata, Rectus Femoris, Iliopsoas and Sartorius. Electromyographic activity was recorded with bipolar 50μ wire electrodes inserted into the respective muscles with a 20-gauge hypodermic needle using Basmajian's single needle technique [32]. Electrode placement was performed using standard electrode insertion sites [33]. Electrode placement was confirmed by electrical stimulation of the muscle through the indwelling electrode, and by voluntary muscle contraction. The EMG system bandwidth was 40–1000 Hz with an overall gain of 1000 [34] (Model MA-300, Motion Laboratory Systems, Baton Rouge, LA, USA). Normalization of the EMG signal was required as electrode insertion was a blind procedure as to the number of muscle fibers sampled. A maximum voluntary contraction was performed, and used to normalize the EMG signal during gait [35]. The limb segments were positioned to allow for maximum isolation of muscle

activity in the respective muscle tests while electromyographic activity (in mV) was recorded [36]. One hundred percent maximum voluntary contraction (100%MVC) was defined as the 1-s interval with the highest mean intensity during a 5-s manual muscle test [37]. Footswitch data (*above*) was used to define gait cycle intervals and foot support patterns. EMG and footswitch data were collected digitally at a rate of 2500 samples/s [37]. After completing the walking trials, another maximum voluntary contraction was recorded to insure continued integrity of the electrode insertion. The digitized EMG data were rectified and integrated [38]. The integrated signal was then corrected for baseline noise. Stride data were divided into 1% GC intervals with %MVC determined at each percent gait cycle interval [37]. Identical processing steps were used for each muscle tested. Linear envelopes (LE) were generated from the rectified, integrated EMG signal for each stride, and were expressed as %MVC at each percent gait cycle interval between onset and cessation times [38]. The within-subject profile for each muscle reflected that muscle's EMG relative intensity (as %MVC) per percent gait cycle interval [39]. The next step involved filtering the within-subject EMG profiles to account for the temporal lag between electrical activity and force production (electromechanical delay, EMD). The onset of electrical activity generally precedes force production. Similarly, the electrical activity returns to baseline levels prior to termination of force in the in-series tendon. The time to peak tension was typically less than the $\frac{1}{2}$-relaxation time [40]–[42]. Peak-to-peak lag across all tested muscles was 66 $\pm$ 8 ms. Electromyogram processing involved a transfer function similar to that relating the EMG signal to a muscle twitch. The input was the within-subject EMG profile for the muscle, whereas the output of the filter mimicked the summation of all twitch waveforms and was depicted as a relative tension waveform for that muscle [43]. A second-order critically-damped Butterworth low-pass digital filter was used to associate EMG input to force output [12]. The post-processed EMG profile $\varepsilon_i(t)$ for each tested muscle was defined by

$$\varepsilon_i(t) = \gamma\,[k_1 E_i(t) + k_2 E_i(t-1) + k_3 E_i(t-2) + K_4 \varepsilon_i(t-1) + k_5 \varepsilon_i(t-2)] \quad (1)$$

where (1) γ represents the ratio $E_i^{max}/\varepsilon_i^{max}$ for that muscle, (*2*) E_i the %MVC for that muscle's EMG profile, (*3*) ε_i the %MVC for the tested muscle's EMG profile following filtering, and (*4*) the values of the constants k_1-k_5 were derived from the ratio of the sampling frequency and the cutoff frequency [44], [45]. The resultant EMG profiles were the neural drive input to the force processor. The net active EFP hip moment was obtained by summing the individual EFP muscle moments

$$M_{active}^{EFP} = \sum_{i=1}^{i=14} M_i^{MTU}. \quad (2)$$

A. Passive Hip Moment

One goal of this research was to assess if a generic neuromuscular model—scaled to each study participant's unique physical dimensions—could be easily used in a clinical environment to estimate in vivo muscle forces. Thus direct measurement of the soft tissue moment contribution was not practical. Hence, best estimates of non-contractile moments were derived from the literature, and applied to the EFP model. Non-contractile tissues create moments that generally restrain movement. Passive hip moments are reported for motion in the sagittal plane (only) [46]–[49]. While the joint range of motion—and resulting passive moment—is likely greatest in the sagittal plane, joint excursions in the frontal and transverse planes can also add to the passive hip moment. The passive component of the hip moment can be substantial, particularly at the end range of movements [50]. These passive moments are insensitive to flexion and extension rates [46], [47], assuring their applicability to this model. For the aforementioned reasons (plus failure to account for hysteresis) the published passive hip moment values may be conservative. Thus an a priori decision was made to select a (liberal) passive hip moment from the reported literature values at each gait-associated (sagittal plane) joint position that might reasonably approximate the net (3D) passive moment. The net EFP moment (M^{EFP}) was defined as the sum at each percent gait cycle of the active and passive moments

$$M^{EFP} = M_{active}^{EFP} + M_{possive}^{EFP}. \quad (3)$$

Inverse dynamics (kinetics plus kinematics) hip moments for each study participant were calculated at each percent gait cycle interval, and mean inverse dynamics hip moments determined. The net EFP and KIN hip moments were determined for each study participant, and then the mean magnitude for each were statistically compared at each percent gait cycle interval

$$\sum_{i=1}^{N} M^{EFP} \approx \sum_{i=1}^{N} M^{KIN}. \quad (4)$$

Statistics

Individual study participant EFP hip moment (3D) magnitudes were determined at each %GC interval, then data pooled across study participants. Mean EFP hip moments (resultant active magnitude plus passive components) were compared with mean KIN hip moment magnitudes at %GC intervals. Inspection of our pilot data indicated that the Model and experimental data satisfy parametric statistical assumptions. Accordingly the coefficient of determination (r^2) was used to index the equivalence of the two hip moment magnitudes (EFP, KIN) [51]. An $r^2 > 0.80$ was required for EFP Model acceptance. This number represents a high level of equivalence between the Model and experimental values, and has been used as an effective benchmark for validation of other muscle models [14].

RESULTS

There was a high degree of concordance between EFP and KIN hip moments ($r^2 = 0.91$, Fig. 1). This correlation is significant at the 0.001 level (*2- tailed*). There was a mean difference of 8 Nm with a range of 0–33 Nm. Of the fourteen muscles examined the Gluteus Medius produced the highest estimated hip gait force ($\approx$ 390N, Fig. 2). When compared to hip abductors the hip extensors produced less

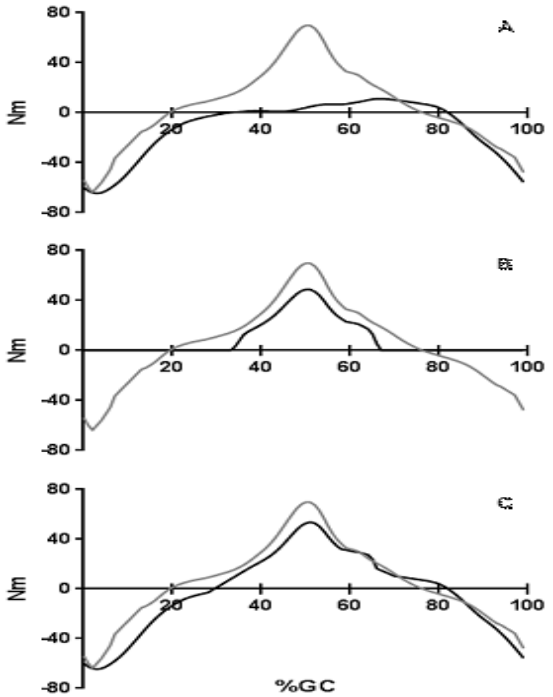

Fig. 1. (A) Mean KIN (*light line*) and active EFP (*dark line*) sagittal plane hip moments are shown. (B) Mean KIN (*light line*) and passive EFP (*dark line*) sagittal plane hip moments are shown. (C) Mean KIN (*light line*) and active plus passive EFP (*net, dark line*) sagittal plane hip moments are shown. Flexion moments are positive.

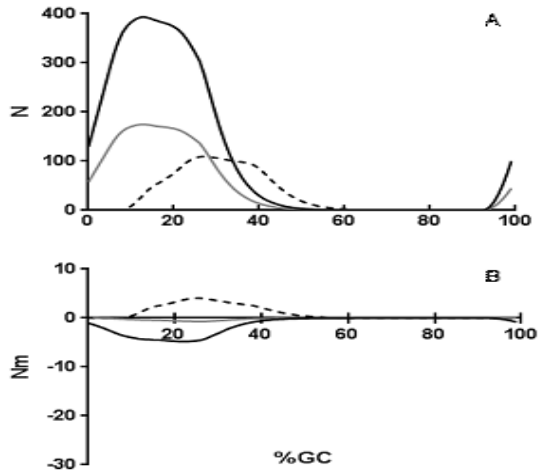

Fig. 2. (A) Mean EFP force estimates for Gluteus Medius (*dark line*), Gluteus Minimum (*light line*) and Tensor Faciae Latae (*dashed line*) are shown. (B) Their corresponding contribution to the sagittal plane active EFP moment are shown. Flexion moments are positive.

estimated force per %GC but generally over a longer duration (Fig. 3). Peak estimated adductor forces were substantially less than predicted values for either hip extensorsor hip abductors. Late stance through early swing showed estimated adductor forces of nearly 135 N (Adductor Longus) and 95 N (Adductor Brevis). Lowest peak force estimates were observed in the hip flexors (130N and 80N for the Rectus Femoris and Iliopsoas, respectively). The passive hip moment was prominent from late single support through early swing [34%–68%GC, Fig. 1(B)].

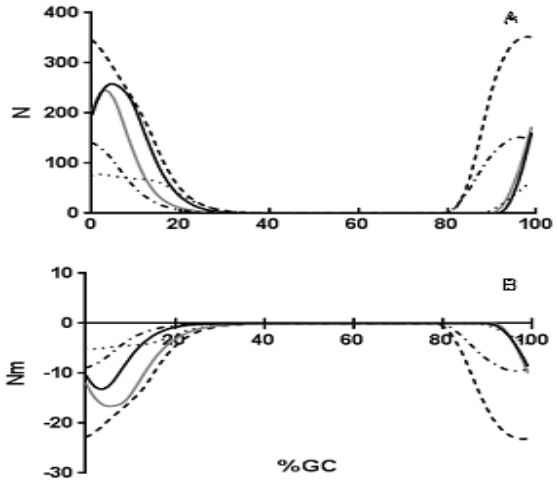

Fig. 3. (A) Mean EFP force estimates for the Semimembranosis (*dashed line*) Semitendinosis (*atternating line*), Long Head of the Biceps Fernoris (*dotted line*), Gluteus Maximus (*solid, dark line*) and Adductor Magnus (*solid, light line*) are shown. (B) Their corresponding contribution to the sagittal plane active EFP moment are shown. Flexion moments are positive.

Fig. 1. (A) Mean KIN (light line) and active EFP (dark line) sagittal plane hip moments are shown. (B) Mean KIN (light line) and passive EFP (dark line) sagittal plane hip moments are shown. (C) Mean KIN (light line) and active plus passive EFP (net, dark line) sagittal plane hip moments are shown. Flexion moments are positive.

DISCUSSION

Comparison of EFP and KIN Moments

A comparison of directly-measured muscle forces (e.g., via strain gauge) with the EMG-to-force processing model estimates would—in theory—be the best way to validate this model. However, both ethical and methodological considerations prevent direct measurement of in vivo muscle forces at the human hip. Thus joint moments are often used as a proxy to estimate these forces. Commonly-used inverse dynamics techniques (kinetics plus kinematics, KIN) synthesize limb segment motion, body segment parameters and externally-applied forces (e.g., ground reaction force) to determine the residual joint moment. In contrast, the EFP hip moment was the summed product of all muscle forces and their respective moment arms, plus the passive hip moment.

Muscles Tested

A substantial number of hip muscles were included in the model, yet a few muscles were excluded from analysis. No gait EMG data was obtained from the Inferior Gemellus, Superior Gemellus, Obturator and Piriformis muscles. These four small hip rotators are not typically studied in a clinical gait analysis environment. Due to their small physiological cross-sectional area, and also short moment arms, their individual and collective contribution to the net hip moment was negligible. Thus little change in the net hip EFP moment was expected as a result of their omission in this analysis. While EMG was not acquired from these four deep hip muscles, other deep muscles were tested. Hence intramuscular electrodes were used to assess the EMG signal of these muscles (e.g., Iliopsoas). Understanding the role of these muscles is important, as many

of them are involved in abnormal walking patterns. Thus, while technically challenging to acquire, intramuscular recordings were essential. However, data obtained with intramuscular electrodes has not gained wide- acceptance. In part, this is due to the need for highly-trained individuals to acquire the signal, and to a lesser extent to minimal discomfort to the patient during testing. An additional concern has been extrapolating the results from the small muscle volume examined with wire electrodes to the entire muscle. Yet the normalized intramuscular (wire) EMG signal correctly reflects the electrical activity of the entire muscle [52]. While surface electrodes could be used to assess more superficial muscles, a single electrode type (intramuscular electrodes) was used to examine all tested hip muscles in order to eliminate a potential confound.

An early stage in model development was the need to define "a muscle", which for these purposes included (A) all muscle fibers bounded by the sarcolemma and (B) the associated proximal and distal tendon structures. The concept of "lumpers versus splitters" was raised as early as 1857 by Darwin [19], and this study has elements of each. Each muscle was considered to be a single force- producing entity ("lumping") and examined as a distinctive entity rather than combined with other synergists ("splitting"). Many of the tested muscles have more than one possible role (i.e., combined hip flexion and external rotation). Muscles with more than one role could, in theory, be further partitioned into functional segments within the muscle. Division of each muscle into smaller, functional components was considered, but ultimately rejected. This decision likely impacted analysis of Adductor Magnus predicted force values more than any other hip muscle, as it may have four distinct anatomic and functional regions [33]. The most proximal aspect of the muscle has been associated with hip joint stabilization, while the remainder (vast majority) of the muscle may displace the thigh through a large range of motion. More commonly Adductor Magnus is considered to have two distinct regions– an adductor portion and a portion that is functionally associated with the other hip extensors [53]. The adductor portion has shorter, more horizontally-oriented muscle fibers. In contrast, the extensor portion has longer, more vertically-oriented fibers. This unique muscle is also dually-innervated, which only makes the decision to model it as one- or several-muscles more complex. During preliminary trials dynamic EMG was acquired with intramuscular electrodes at both the adductor and extensor regions. Neither EMG acquisition site had electrical activity during the initial- and mid-swing phases. Thus the electrical response of Adductor Magnus (and it follows, force production) was clearly out-of-phase with other hip adductors. EMG acquired from the posterior aspect of the muscle was notable for a higher-amplitude signal and more consistent activity across-individuals during the preliminary trials. As a result the more posteriorly- located site was used to obtain Adductor Magnus EMG in all study participants. Consequently its contribution to predicted hip extensor forces (and moments) may be slightly overestimated, as its muscle volume included both the hip extensor and adductor aspects. Recent work suggests that two hip abductors (Gluteus Medius and Minimus) may in fact be two or more distinct functional units, rather than a homogeneous muscle. Semciw reports

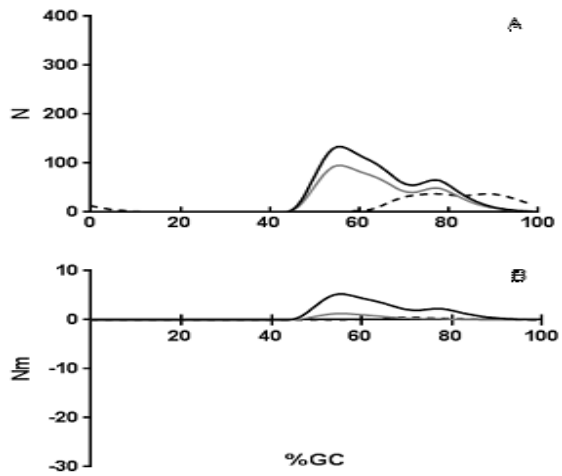

Fig. 4. (A) Mean EFP force estimates for Adductor Longus (*dark line*), Adductor Brevis (*light line*) and Gracilis (*dashed line*) are shown. (B) Their corresponding contribution to the sagittal plane active EFP moment are shown. Flexion moments are positive.

that Gluteus Medius has three distinct anatomic segments (posterior, lateral, and anterior) [54]. Further, Gluteus Minimus is noted to have separate anterior (vertically-oriented fibers) and posterior (horizontally-oriented fibers) segments [55]. This study's EMG insertion sites were at the lateral aspect of Gluteus Medius and the anterior aspect of Gluteus Minimus. No gait data was presented for the individual Gluteus Medius segments. Normalized gait EMG profiles (using intramuscular electrodes with ultrasound guidance) were presented for Gluteus Minimus [55]. While there was variability in peak relative intensity for the anterior and posterior portions of this muscle, the across-subject EMG profiles were remarkably similar with respect to onset and cessation times plus timing of peak amplitude. As there is scant evidence of marked differences in gait EMG for either muscle, the presumption that each of the gluteal abductors essentially works as a single contractile unit (with respect to muscle force) seems reasonable. In contrast to the aforementioned hip extensors and abductors, two hip flexors were assessed as a single muscle. The EFP model combined the physiological cross-sectional area of the Iliacus and Psoas ("Iliopsoas"), and used the EMG obtained for that muscle's force and moment production. Looking ahead, future EMG-to-force processing models might examine the respective contribution of selected aspects of these "muscles within muscles", with a goal of improving post-surgical outcomes and to better target rehabilitation programs.

The timing and relative intensity of muscle activity shown here were consistent with previously- published findings [53]. A surprising finding was that the estimated peak force production of these relatively powerful muscles was quite modest. The largest estimated hip muscle force was slightly less than 400N, generated by Gluteus Medius during weight acceptance. While the Gluteus Maximus has substantially larger physiological cross-sectional area [20], its estimated peak force during walking was smaller (Fig. 3). In fact, only four hip muscles had estimated peak gait forces that approachedor exceeded 250N (Fig. 4).

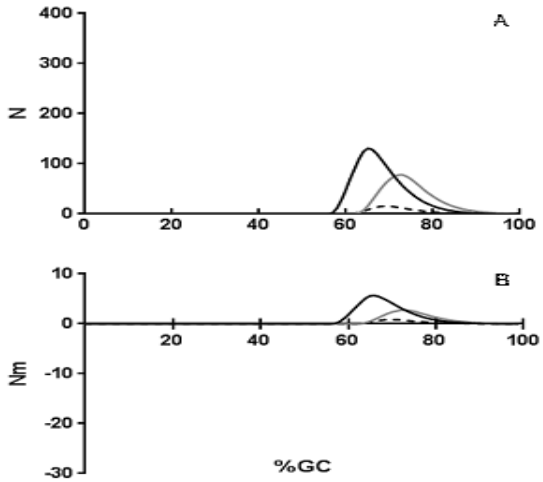

Fig. 5. (A) Mean EFP force estimates for Rectus Femoris (*dark line*), Iliopsoas (*light line*) and Sartorius (*dashed line*) are shown. (B) Their corresponding contribution to the sagittal plane active EFP moment are shown. Flexion moments are positive.

The maximal estimated force produced by any hip muscle was well below that predicted for either the lateral and medial heads of the Gastrocnemius (350N and 750N, respectively) [14] or the Soleus during self-selected-speed normal adult walking (1950N) [14], [56], [57] (Fig. 5). These findings have implications for current rehabilitation strategies. Despite comparatively modest muscle cross-sectional areas the hip abductors exhibited relatively high estimated peak muscle forces. Recent work has shown that hip abductor muscle weakness substantially increased muscle cost during gait [36], and that these muscles may produce gait moments that approach 40% of capacity [58]. That is, they have a relatively small factor of safety, and even modest loss of muscle strength can adversely affect walking ability. Additionally distal muscle force production is substantially greater than forces produced by more proximal lower extremity muscles in normal adult gait [14]. Further strength loss after neurological injury or disease (e.g.,. stroke) preferentially reduces distal more than proximal muscles [59].

Hence it is interesting to note that current robotic- assisted gait training devices tend to focus on restoration of hip extensor and knee extensor moments (as noted by actuator placement and capability), while weakness of hip abductors and plantar flexors may be clinically more important for achieving more efficient gait.

This suggests that a reexamination of current robotic-assist strategies is needed if restoration of efficient gait is to be achieved using these devices.

Potential Error Sources

One goal of this research was to examine if an approach that applied a study participant's EMG input and anthropometry to a generic neuromuscular model would yield reasonable estimates of in vivo muscle forces. The results presented (similarity in shape and magnitude of the KIN and EFP hip moment curves) suggests that this goal was met, and further that individual muscle force estimates are reasonable. However, there are several potential sources of "error" in any model, and this approach in no exception. First, the statistical test used does not directly evaluate closeness of fit of moment amplitudes. The coefficient of determination (r^2) measures how closely two variables are correlated, yet a high r^2 value does not guarantee that the data sets are essentially identical (which is essential for model acceptance). While not shown here, it is possible to present large differences in KIN and EFP hip moment amplitudes, yet attaining r^2 values if those moment curves have highly similar shapes (but not amplitude). Visual inspection of the data [Fig. 1(C)] clearly shows that the EFP and KIN hip moments were both closely correlated *and* favorably similar in magnitude. The method of estimating muscle forces required precision in obtaining a true maximum voluntary contraction (MVC). EFP muscle forces were scaled by MVC, and less-than-full effort during muscle testing could result in overestimates of gait muscle forces. To minimize this possibility the hip was placed in a neutral position [36] during all maximum muscle tests. In each maximum muscle test hip motion was resisted through the 5-s muscle test interval. Hence the possibility of less-than-maximal contractions on muscle testing was lessened, and the likelihood of muscle testing substantially affecting force approximations was reduced.

The magnitude of the moment arm estimate was another potential error source. Variability in moment arm magnitude ranges (scaled model versus MR imaging) from 1–2 cm, and thus individual muscle moment estimates errors could be greater than 5 Nm [60]. Despite the potential for moment arm-induced error Correa et al [60] found that predictions of muscle function were essentially unaffected. It is unlikely that every muscle moment arm magnitude in every tested individual (20 study participants, 14 muscles examined per participant = 280 unique moment arm estimates) was spot-on. However, visual inspection of each study participant's EFP hip moment with their corresponding KIN hip moment showed that the moment estimates were highly similar, particularly during periods with little or no passive moment contribution. While it is conceivable that multiple offsetting errors in moment arms magnitudes and muscle force estimates could yield similar KIN and EFP comparisons, a more likely conclusion is that there were no systematic errors in the moment arm magnitudes selected. Further, a similar EFP approach in estimating ankle moments (and muscle forces) exhibited similar close correlations between EFP and KIN moments [14]. Thus the disparities between model (EFP) and experimental results (KIN) should principally be due to errors in estimates of muscle force when the muscle was demonstrated to be active. There was a substantial contribution to the EFP net hip moment by passive tissues. There are at least two potential errors associated with the method used to estimate the passive hip moment contribution. First, it is highly unlikely that the moment contribution of passive (non-contractile) tissues–based on values obtained from the literature– exactly matched the passive hip moment (at each %GC interval) for any tested individual.

Without direct measurement of each study participant's passive hip moment (in each plane) there is uncertainty as to the precision of these estimates. In addition, previous published work has only identified the passive hip moment across a range of sagittal plane joint angles observed in gait.

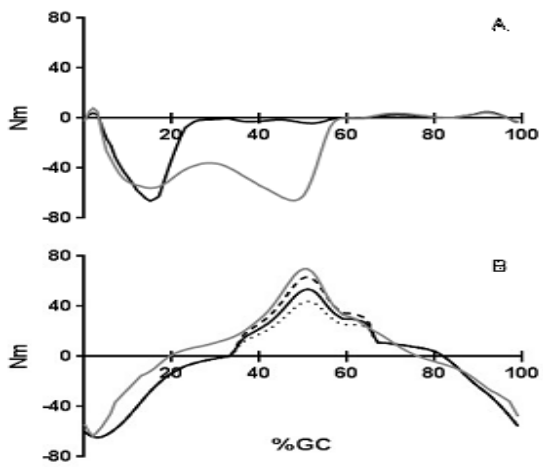

Fig. 6. (A) Mean sagittal plane KIN (*light line*) and EFP (*dark line*) hip moments are shown. Adduction moments are positive. (B) Mean KIN (*light line*) and net EFP sagittal plane moments are shown. EFP moments are shown. EFP moments presented are 100% of the a priori magnitude as determined by joint position (*dark line*), for the passive moment increased by 20% (*dashed line*), and reduced by 20% (*dotted line*) from a priori values. Flexion moments are positive.

Clearly the soft tissues also provide a contribution to the net moment in the frontal and transverse planes (particularly at the end ranges of motion), yet these values were not available [Fig. 6(A)]. However, direct measurement of passive moments is at odds with one goal of this research. That is, applying EMG input to a scaled, generic model for use in a clinical environment. It was obvious prior to commencing the research that direct measurement of passive moments was not clinically practical. An unanticipated finding was that this step was also unnecessary for this application. Previous EMG-to-force processing models have ignored the moment contribution by non-contractile tissues [14], [61], and this omission did not dramatically affect the precision of their models. However those studies examined forces produced by ankle muscles, and ignoring the passive ankle moment did not adversely affect model acceptance in each case. In this investigation failing to account for the passive moment would have resulted in wide disparity of the KIN and EFP hip moments [Fig. 1(A)], and the EFP hip muscle force estimates would have been falsely rejected. Thus the inclusion of the hip passive moment (even with a sub-optimal degree of precision) minimized this unwanted possibility. Post-hoc analyses suggested that the net in vivo passive moment contribution at the hip might be even greater than predicted with the model. Post hoc analyses showed that passive hip moments 120% of values selected a priori lead to an even closer correlation of KIN and EFP net moments [Fig. 6(B)]. Even passive hip moments that were 80% of selected values would have resulted in model acceptance. The (possible) presence of muscle- associated series elastic and parallel elastic components of the Hill-based model in the passive hip moment calculations was also a potential error source. Inclusion of these muscle-related non-contractile tissues could theoretically influence the passive hip moment. As noted above the passive hip moment contribution only influences acceptance or rejection of the model, and has no bearing on the magnitude or timing of the hip muscle force estimates. Further, this potential confound would be limited to those epochs where both active muscle contraction and passive moments were present. This potential error source was serendipitously minimized as EFP active moments were low magnitude when EFP passive moments were high. That is, passive and active EFP moments were for the most part out-of-phase with each other. The passive moment contribution lasted only about one- third of the gait cycle (34%–68%GC [Fig. 1(B)], and until the onset of active moment contributions by the primary hip flexors (Rectus Femoris onset at 57%GC [Fig. 5(A)] only non-contractive tissues could account for the large hip moment observed. Visual inspection of the respective moment curves indicated that there was little divergence between EFP and KIN moments during periods where muscle forces were predicted to be large [Fig. 1(A)].

Future Directions

Estimates of in vivo muscle forces for neurologically-intact adults performing low-velocity movements (comfortable-speed walking) were presented here. Yet extrapolation to other types of activity (e.g., ballistic movements) or populations (e.g., post-stroke individuals) should be done with great caution. Some implications on the role of the hip muscles can be based on muscle forces, alone. However, a more complete understanding would require assessment of the power and work produced by these same muscles. Clearly further research is necessary to enhance our understanding of human walking.

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